1.8. quadratische Gleichungen

PQ-Formel

$x^2 + px - q = 0$ hat Lösungen $x_{1/2} = -p/2 \pm \sqrt{(p/2) - q}$

ODER:

ABC-Formel

$ax^2 + bx + c = 0$ hat Lösungen $x_{1/2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(a) $x^2 - x - 1 = 0$ hat die Lösungen

$$x_{1/2} = \frac{1}{2} \pm \sqrt{\frac{1}{4} + 1} = \frac{1}{2} \pm \sqrt{\frac{5}{4}} = \frac{1}{2} \pm \frac{\sqrt{5}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

(b) $a^2x^2 + 2x = 1 \quad \Rightarrow \quad a^2x^2 + 2x - 1 = 0$

hat die Lösungen

$$x_{1/2} = \frac{-2 \pm \sqrt{4 + 4a^2}}{2a^2} = \frac{-2 \pm 2\sqrt{1 + a^2}}{2a^2}$$

$$= \frac{2 \cdot (-1 \pm \sqrt{1 + a^2})}{2 \cdot a^2} = \frac{-1 \pm \sqrt{1 + a^2}}{a^2}$$

(c) $x^4 - 6x^2 + 8 = 0 \xrightarrow{z = x^2 \text{ "Substitution"}} z^2 - 6z + 8 = 0$

hat die Lösungen

$$z_{1/2} = \frac{6}{2} \pm \sqrt{\left(\frac{6}{2}\right)^2 - 8} = 3 \pm \sqrt{9 - 8} = 3 \pm 1$$

$$\Rightarrow \boxed{z_1 = 4 \ \vee \ z_2 = 2}$$

Rücksubstitution: $x^2_{1/2/3/4} = z_{1/2}$

$$\Rightarrow \boxed{x_1 = 2 \ \vee \ x_2 = -2 \ \vee \ x_3 = \sqrt{2} \ \vee \ x_4 = -\sqrt{2}}$$